SAHAYI

WhatsApp-Based AI Voice Assistance, Caregiver Coordination & SOS Escalation

Complete product specification

A Malayalam-first voice assistant for senior citizens and persons with disabilities. Sahayi receives WhatsApp voice messages, transcribes them, understands the user's reported need using authorized profile context, keeps connected members informed, and escalates potential emergencies through a defined SOS workflow.

The assistant actively converses with the user, asks short clarifying questions, and can raise the alert level as the conversation develops.

Core promise: A person should be able to ask for help naturally through WhatsApp voice messages, while trusted contacts are informed and can coordinate a response.

Prepared for: BIA Hackathon

Product concept • User journeys • Conversation and SOS logic • Architecture • Data model • Implementation plan • Safety constraints

1. Product Overview

Objective

Build a WhatsApp-integrated, Malayalam-first assistant that supports senior citizens and persons with disabilities during everyday difficulties and possible emergencies. The system should converse with the user, understand what they are reporting, consult relevant authorized profile information, and coordinate with connected family members or caregivers.

Key product behaviours

- The user can send a Malayalam or English voice message through WhatsApp.
- Sahayi responds and asks simple, relevant follow-up questions instead of treating the first transcript as the complete story.
- Connected members receive an initial notification that the user has contacted Sahayi and that an interaction is in progress, subject to consent and notification settings.
- The system monitors the conversation for signs that may indicate an urgent or emergency situation.
- If the situation appears to be an emergency, Sahayi sends an SOS-level alert to connected members.
- A connected member can use the WhatsApp bot's SOS action to request escalation and initiate the supported emergency-calling workflow.
- The system uses the user's preset location and relevant profile details to prepare the escalation information.

2. Users and Connected Members

Role	Capabilities
Senior citizen / person receiving assistance	Sends voice messages; answers Sahayi's questions; receives simple instructions and status updates.
Primary caregiver / family member	Receives activity and SOS alerts; reviews the situation; can initiate the SOS escalation workflow.
Other connected member	Receives notifications according to permissions; can respond or acknowledge an alert.
Authorized profile manager	Maintains patient profile, emergency contacts, preferred language and preset location.

The product must support consent-based contact linking. A person should know which contacts are connected and what information may be shared with them.

3. WhatsApp Voice Intake and Speech-to-Text

3.1 Incoming voice message

1. User sends a WhatsApp voice message to the Sahayi business number.
2. WhatsApp Business Cloud API sends a webhook event to the FastAPI backend.
3. Backend validates the webhook and sender, retrieves the audio media, and associates the message with a registered profile.
4. Audio is passed to language detection and then routed to the matching STT provider.

3.2 Language routing

Detected language	STT provider	Expected output
Malayalam	Sarvam AI Speech-to-Text	Malayalam transcript
English	Groq Whisper	English transcript
Uncertain / mixed	Retry, alternate route or ask user to repeat slowly	Transcript marked uncertain until verified

The first MVP supports Malayalam and English. Mixed-language messages should not be silently misclassified. The assistant should ask the user to repeat or confirm uncertain words when they materially affect the interpretation.

4. Conversational AI Agent

Sahayi is not only a transcription pipeline. It should conduct a short, context-aware conversation to understand what the user needs.

4.1 Agent responsibilities

- Interpret the transcript and identify the user's stated need or difficulty.
- Ask one short, plain-language follow-up question at a time when essential information is missing.
- Confirm critical details such as what happened, whether the user is alone, whether they can speak or move safely, and whether help is needed now.
- Use the authorized patient profile to add relevant context without assuming that historical conditions explain the current event.
- Classify the interaction using a predefined escalation policy.
- Generate a concise summary for connected members and maintain a record of the conversation and alert state.
- Avoid lengthy or complex questions, especially when the user may be distressed.

4.2 Example interaction

User voice message: “□□□□□□□ □□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□. □□□□□□□□□□□□□□ □□□□□□□□□□□□□.”

Transcript: The user reports sudden dizziness and difficulty standing.

Sahayi: “□□□□□□ □□□□□□ □□□□□□□□□□□□□? □□□□□□□ □□□□□□□□□□ □□□□□?” (Are you alone right now? Is anyone nearby?)

Connected member update: “Adi's registered contact has started a conversation with Sahayi. The assistant is checking what help is needed.”

Agent behaviour: The assistant continues to ask only necessary questions, checks the profile for relevant context, and escalates according to the safety policy.

This is an illustrative interaction, not a medical diagnosis or a validated triage protocol.

5. Connected-Member Notifications and SOS Levels

Connected members should be kept informed during the conversation. The system should distinguish routine activity updates from urgent SOS escalation.

Level	Trigger / meaning	System action
Level 0 — Conversation started	User has contacted Sahayi; no emergency signal identified yet.	Send a discreet start-of-conversation notification to configured connected members, subject to user consent and settings.
Level 1 — Attention	User reports a difficulty that may need assistance, but available information does not indicate an immediate emergency.	Continue conversation; send an appropriate status update if configured; request clarification.
Level 2 — Urgent	Reported facts suggest the user may need prompt human assistance.	Send an urgent alert to designated members with the reported facts and requested action.
Level 3 — SOS	Conversation contains predefined high-risk indicators or a member manually raises SOS.	Send an SOS-level alert to connected members; begin the configured escalation workflow immediately.

Escalation principle: Do not wait for the conversation to finish if the user appears to be in immediate danger. The system can ask short questions while notifying contacts. If the user cannot respond, follow the predefined safety protocol rather than assuming they are safe.

6. Member-Initiated SOS from WhatsApp

Connected members receive WhatsApp notifications containing a clear action to escalate the situation. A member who believes the situation is dangerous can select an SOS action in the WhatsApp interaction.

6.1 Proposed workflow

1. A connected member receives an SOS or urgent notification with a concise situation summary.
2. The member selects the “Initiate SOS” action in the WhatsApp bot.
3. The backend authenticates the member and verifies that they are authorized to initiate escalation for this profile.
4. The bot presents the registered location and asks the member to confirm that it is current and appropriate for dispatch.
5. The system invokes the configured, supported emergency-calling or escalation channel.
6. The member is shown whether the call/escalation was successfully initiated, failed, or requires manual action.
7. The incident record stores the time, initiating member, location used and outcome.

6.2 Preset location

Each profile can store a primary assistance location, such as the senior's home address, plus optional location instructions (building, floor, apartment, landmark, access notes). The location must be confirmed and periodically reviewed. If the user is away from the preset location, the system must not represent the home address as their current location.

6.3 Critical WhatsApp / emergency-call limitation

The WhatsApp Business Cloud API is a messaging interface; it does not, by itself, place a normal telephone call to India's emergency services or guarantee that responders will be dispatched to a preset address.

For the prototype, a WhatsApp SOS action can alert connected members and open a supported call action on the member's device (for example, a clearly labelled call-emergency-services link where supported), or connect to a separately configured telephony / emergency-response integration. Any automated calling integration must be validated with the relevant emergency-service operator, telecom provider, local regulations and operational procedures before being presented as a real dispatch capability.

In India, 112 is the national emergency response number. The prototype must not claim that an API call to 112 or a WhatsApp button automatically transmits the patient's profile or location unless that capability has been formally confirmed and tested with the authorized service. If the integration is unavailable, the bot must clearly instruct the member to call 112 directly and communicate the verified location.

6.4 Failure handling

- If automated calling is unavailable, immediately show the member a manual call instruction and the verified address.
- If location is missing, stale or uncertain, flag it prominently and ask the member to verify it.
- If the call action fails, do not display a successful status; preserve the alert and offer the manual fallback.
- Do not allow an LLM to decide whether emergency services should be called without deterministic policy checks and human/authorized escalation rules.

7. Patient Profile and History Dashboard

An authorized caregiver or profile manager can register and maintain a profile.

Profile category	Fields / examples
Personal details	Name, age, preferred language, profile identifier.
Medical context	User-provided medical history relevant to assistance; source and last-updated date.
Medication and allergies	Current medication information and known allergies, if supplied.

Profile category	Fields / examples
Emergency contacts	Name, relationship, verified phone number, notification permissions and escalation order.
Accessibility needs	Mobility, hearing, vision and communication preferences.
Preset location	Address, landmark, building/floor details, last verified date and optional access instructions.
Consent and preferences	Consent status, data-sharing preferences, notification rules and emergency contacts approved by the user.

Only the minimum relevant profile context should be included in an alert. Medical history must not be broadcast to every connected member by default.

8. Complete System Architecture

Component	Responsibility
WhatsApp Business Cloud API	Receives voice messages and sends replies / member notifications.
FastAPI webhook	Validates incoming events, maps sender to account and retrieves media.
Language detection	Routes Malayalam to Sarvam and English to Groq Whisper.
STT services	Return transcript and transcription metadata.
Conversation manager	Maintains conversation state and asks follow-up questions.
Problem / urgency analyzer	Produces structured facts and escalation indicators using a constrained policy.
Patient profile service	Retrieves authorized profile, contact and location information.
SOS orchestration service	Handles Level 0–3 state, member confirmation, deduplication and escalation.
Calling / emergency integration	Uses only a validated, supported channel; otherwise provides a manual call fallback.
Supabase PostgreSQL	Stores profiles, permissions, conversation summaries and incident records.
React dashboard	Manages profiles, contacts, location, consent and incident history.

9. Suggested Backend Structure

```
backend/ app/ main.py whatsapp/ webhook.py media.py sender.py interactive_actions.py speech/ language_detector.py
sarvam_stt.py groq_stt.py conversation/ manager.py session_store.py question_planner.py agents/ problem_analyzer.py
urgency_classifier.py message_generator.py patients/ profile_service.py history_service.py location_service.py
notifications/ member_alerts.py sos_orchestrator.py emergency_call_adapter.py database/ supabase.py safety/
escalation_policy.py consent.py audit_log.py tests/ requirements.txt .env.example README.md
```

10. Suggested API Endpoints

Endpoint	Purpose
POST /webhooks/whatsapp	Receive and validate WhatsApp webhook events.
POST /api/conversations/{id}/process	Process a transcript and advance the conversation.
GET /api/patients/{id}	Retrieve an authorized patient profile.
POST /api/patients	Create a patient profile.
POST /api/patients/{id}/contacts	Add / update authorized connected members.
POST /api/incidents/{id}/raise-sos	Raise an SOS under the escalation policy.
POST /api/incidents/{id}/member-sos	Process an authenticated member's SOS action.
POST /api/incidents/{id}/call-emergency	Invoke a validated calling adapter or return manual fallback instructions.
GET /api/incidents/{id}	Retrieve incident status and audit history.
GET /health	Check backend health.

11. Recommended Technology Stack

Layer	Technology	Purpose
Messaging	Meta WhatsApp Cloud API	Voice intake, replies and interactive member actions.
Backend	Python + FastAPI	Webhook, orchestration and APIs.
Malayalam STT	Sarvam AI	Malayalam speech transcription.
English STT	Groq Whisper	English speech transcription.
AI interpretation	Gemini or Groq LLM	Structured intent and problem interpretation.
Database	Supabase PostgreSQL	Profiles, permissions, contacts and incidents.
Frontend	React + Vite	Caregiver and profile dashboard.
Deployment	Render + Vercel	Backend and frontend hosting.
Emergency calling	Validated provider / authorized integration	Only if supported and approved; otherwise manual emergency call fallback.

12. Privacy, Safety and Reliability Requirements

- Verify WhatsApp webhook signatures and validate every inbound event.
- Authenticate connected members before accepting an SOS action.
- Obtain informed consent for profile storage, conversation processing and contact notifications.
- Encrypt sensitive data in transit and at rest; apply role-based access controls.
- Minimize audio and transcript retention; provide a clear retention and deletion policy.
- Prevent duplicate alerts using WhatsApp message IDs, incident IDs and idempotent processing.
- Keep an auditable record of alert level changes, member actions, location confirmation and call outcomes.
- Clearly separate transcript facts, profile facts and AI-generated inferences.
- Never diagnose, prescribe medication or claim a medical emergency has been verified by an LLM.
- Do not imply that Sahayi is continuously monitored or that a message guarantees a human response.

- When immediate danger is suspected, tell the user/member to contact emergency services directly; do not delay while waiting for AI processing.
- Test low-quality audio, silence, language misclassification, unavailable APIs, stale location and failed notification delivery.

13. Four-Hour Hackathon MVP Plan

Priority	Feature	Build target
1	WhatsApp webhook and voice media retrieval	Must have
2	Malayalam Sarvam STT and English Whisper STT	Must have
3	Conversation flow with short follow-up questions	Must have
4	Initial connected-member notification	Must have
5	Patient profile lookup with relevant context	Must have
6	Deterministic alert levels and SOS message	Must have
7	Member-initiated SOS action and verified location display	Must have
8	Actual emergency-service calling integration	Only if a validated integration is available; otherwise demo manual call fallback
9	Full incident history and advanced dashboard	If time permits

For a four-hour prototype, demonstrate the WhatsApp interaction, language-specific transcription, follow-up conversation, initial member notification, SOS escalation and member-initiated SOS action. Do not fake an emergency-services dispatch. Clearly label any call action as a prototype or manual handoff unless the complete integration has been verified.

14. Three-Minute Demonstration Flow

1. 0:00–0:25 — Introduce the challenge: a senior citizen may need help but may find it difficult to navigate an app or explain the situation.
2. 0:25–0:55 — Show the registered profile, connected members and preset home location using fictional data.
3. 0:55–1:25 — Send a Malayalam WhatsApp voice message; show Sarvam transcription and Sahayi's first follow-up question.
4. 1:25–1:55 — Show the initial connected-member notification while the conversation continues.
5. 1:55–2:25 — Demonstrate the agent identifying a high-risk situation and issuing an SOS-level alert.
6. 2:25–2:50 — A connected member selects Initiate SOS; show authentication, location confirmation and the validated call path or manual emergency-call fallback.
7. 2:50–3:00 — Close with the impact: accessible communication, faster caregiver awareness and structured escalation.

15. Acceptance Criteria

- A WhatsApp voice message can be received and securely associated with the correct registered user.
- Malayalam audio is routed to Sarvam AI and English audio to Groq Whisper.
- The bot can ask at least one contextually relevant follow-up question.
- Configured connected members receive the conversation-start notification.
- The system can raise an SOS-level alert based on predefined rules and the conversation.
- An authorized connected member can initiate the SOS workflow through WhatsApp.
- The preset location is displayed with its verification status.

- The emergency-call step reports its actual outcome and provides a manual fallback.
- The dashboard shows the profile and incident status using synthetic demo data.

Final product positioning: Sahayi is a voice-first assistance and caregiver-coordination system. It turns a simple WhatsApp voice message into a guided conversation, keeps trusted contacts informed, and provides a controlled path for escalating urgent situations.